

Basic AWS Concepts:

What is AWS in simple terms?

AWS stands for Amazon Web Services. It is the world's largest cloud computing platform. In simple terms, AWS is like an enormous utility company for computing power. Instead of buying and managing your own physical computer servers, storage, and networking equipment, you rent these services from Amazon over the internet. You only pay for what you use, just like electricity or water.

What is EC2 and why do we use it?

EC2 stands for Elastic Compute Cloud. You can think of an EC2 instance as a Virtual Computer in the cloud.

- What it is: A customizable virtual server (called an instance) that you can start, stop, resize, and fully control. You choose the operating system (Windows, Linux, etc.) and the power (CPU, RAM).
- Why we use it: We use it to run almost any kind of application or workload, like hosting a website, running a database, or performing complex calculations. It's the basic computing power foundation of many apps on AWS.

What is S3 and what makes it different from normal storage?

S3 stands for Simple Storage Service. It is an Object Storage service.

- What it is: A service for storing massive amounts of data as objects (like files: images, videos, backups, documents) in containers called buckets.
- What makes it different from normal storage:
 - Unlimited Capacity: Your regular computer hard drive has a limit, but an S3 bucket has virtually unlimited storage.
 - Durability and Reliability: Data in S3 is automatically copied and spread across multiple locations and facilities to ensure it's almost impossible to lose (designed for 99.999999999% durability). Your local hard drive can fail, but S3 is built to resist catastrophic failures.
 - Accessibility: You can access your files from anywhere in the world over the internet using a simple web address.

What is AWS Lambda and when should you use it?

AWS Lambda is a serverless compute service.

- What it is: It lets you run your code without managing any servers. You upload your code, and AWS automatically handles all the server maintenance, scaling, and capacity planning.
- When you should use it:
 - To run code in response to events (e.g., run a function automatically every time a new image is uploaded to S3).
 - To build an API backend that only runs when a user makes a request (Pay-per-use, only for the milliseconds your code executes).
 - For automating tasks like cleanup jobs or sending notifications on a schedule.

What is IAM in AWS and why is it important?

IAM stands for Identity and Access Management.

- What it is: It's the core security service that lets you control who can access which AWS services and resources. It manages users, groups, and permissions.
- Why it is important: It is the gatekeeper for your cloud account. It allows you to follow the principle of least privilege, meaning you can give a user or an application only the minimum permissions necessary to do its job, which is a critical security practice to prevent accidental or malicious damage.

AWS Scenario-Based Questions:

Your application needs to store user images safely and cheaply. Which AWS service will you choose and why?

- Service: Amazon Simple Storage Service (S3)
- Why: S3 is the best choice for storing unstructured data (like images and videos). It offers high durability (images won't be lost), and it is cost-effective, especially by using S3 Standard-IA (Infrequent Access) or S3 Glacier for older, less-used images.

Your website is getting slow because more users are coming every week. How would you scale the backend on AWS?

You would use a combination of these services to automatically scale:

1. Elastic Load Balancer (ELB): To automatically distribute incoming user traffic across multiple servers.
2. EC2 Auto Scaling: To automatically increase the number of running EC2 servers when the load (traffic) gets high, and decrease them when traffic drops (to save money). This is called Horizontal Scaling.

You want to run a piece of code only when an event happens, without managing servers. Which AWS service fits this use case?

- Service: AWS Lambda
- Why: Lambda is the serverless solution built for event-driven computing. You write the code, and Lambda runs it instantly in response to triggers (the event) without you ever needing to maintain a server.

Your company wants to restrict access so that only specific people can use certain AWS services. How would you implement this?

- Service: AWS Identity and Access Management (IAM)
- How: You would use IAM Policies to define specific permissions for each user or group (e.g., "Allow Developer-Group to read from S3 bucket 'my-app-data' but deny them permission to delete anything").